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Title Page

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Abstract

The classical work by Robert C. Hickson showed in 1980 that the addition of a resistance training protocol to a predominantly aerobic program could lead to impaired leg strength adaptations in comparison to a resistance-only training regimen. This interference phenomenon was later highlighted in many reports, including a meta-analysis. However, it seems that the interference effect has not been consistently reported, probably because of the complex interactions between training variables and methodological issues. On the other side of the medal, Dr Hickson and colleagues subsequently (1986) reported that a strength training mesocycle could be beneficial for endurance performance in running and cycling. In recent meta-analyses and review articles, it was demonstrated that such a training strategy could improve middle- and long-distance performance in many disciplines (running, cycling, cross-country skiing and swimming). Interestingly, it appears that improvements in the energy cost of locomotion could be associated with these performance enhancements. Despite these benefits, it was also reported that strength training could represent a detrimental stimulus for endurance performance if an inappropriate training plan has been prepared. Taken together, these observations suggest that coaches and athletes should be careful when concurrent training seems imperative in order to meet the complex physiological requirements of their sport. Therefore, this brief review will present a practical appraisal of concurrent training for sports performance. In addition, recommendations will be provided so that practitioners could adapt their interventions based on the training objectives.

Keywords: strength and aerobic training, neuromuscular performance, middle- and long-distance performance

While pursuing his postdoctoral studies, Robert C. Hickson gradually included some running sessions to his habitual strength-training program. This anecdote later led to a research project, which became a seminal paper in the field of concurrent training ². The 10-week protocol ³ included one aerobic training (AT) group, one resistance training (RT) intervention and a concurrent training (CT) group, which completed all AT and RT sessions. Participants in the RT group (n=8) had to complete a lower body maximal strength training protocol (3-5 sets of 5 repetitions at approximately 80% of 1RM) 5 days per week. In the AT intervention (n=8), the program consisted of 3 weekly interval training sessions to develop maximal aerobic

While the first obvious flaw of this study relates to the training load, which was dramatically increased in the CT group, potentially leading to an important fatigue state compromising potential adaptations, later studies revealed that the addition of AT to a RT mesocycle could be accompanied with setbacks in terms of strength and neuromuscular power adaptations. Indeed, a meta-analysis including 21 studies revealed that lower body neuromuscular power was particularly compromised after such a training regimen ⁶. Interestingly, it was reported that when AT consisted of running exercises, the interference was more pronounced in comparison to cycling activities. The authors speculated that these results could be due to the important eccentric phase involved in running, potentially increasing muscle damage and reducing strength and power adaptations. In addition, a dose-response relationship was described, with AT duration and frequency negatively associated with neuromuscular performance.

In order to explain this interference phenomenon, both chronic and acute hypotheses have been suggested ⁷. While the acute hypothesis referred to the residual fatigue induced by AT leading to a reduced ability to develop tension during the strength-training component, the chronic hypothesis stated that the interference was due to different physiological adaptations (e.g. muscle activation, muscle fiber type, hypertrophy, glycogen stores depletion) elicited by both training modalities. Indeed, AT and RT are typically associated with two extremes of an effort duration/energy metabolism continuum ². From a muscle fiber perspective, the complex nature of combined sprint and endurance performance has been highlighted with the observation of an inverse relationship between muscle fiber size and oxidative capacity ¹. At the molecular level, it was originally thought that the activation of the adenosine monophosphate kinase (AMPK) by AT could inhibit the mammalian target of rapamycin (mTOR), which is activated by RT ⁸. Consequently, this molecular cascade would then reduce the protein synthesis typically associated with RT ². However, this potential mechanism was discarded considering that such a divergent cascade (AT:AMPK and RT:mTOR) seems too simplistic ⁹ and that the activation of AMPK was reported to have minimal effects on mTOR in humans ¹⁰. While a molecular approach could not fully explain the interference effect observed after CT, insights about the participants' training status were recently added to this discussion. Coffey and Hawley ¹¹ suggested that a CT modality was detrimental mainly in trained individuals looking for more specific exercise adaptations. According to this model, beginners would be less affected by such a CT mesocycle since, at this stage, training adaptations are more general with any increase in physical activity showing a potential to induce large changes in molecular adaptations, no matter the exercise modality ¹¹. In support of this statement, their review highlighted that untrained individuals could improve muscle mass after AT whereas RT could enhance oxidative capacity in beginners ¹¹. Coffey and Hawley also suggested that the extremes found in the effort duration/energy metabolism

continuum (e.g. Olympic weight lifter vs. marathon runner) are only truly observed after many years of specific training. Therefore, these highly trained individuals would need greater and, importantly, more specific training loads to increase the adaptation response, which could make these individuals more vulnerable to CT¹¹. However, this training status hypothesis might be partially questioned considering that no differences between training groups (untrained vs. trained) were observed for strength development during CT⁶. While many questions are still debated, readers interested in molecular adaptation mechanisms and methodological considerations are invited to consult review articles published on the topic^{9 12}.

Contrary to this interference hypothesis, a report published in 2013 suggested that the hypertrophic response was not altered by the addition of AT to a RT mesocycle¹³. Indeed, in this study, 10 moderately trained men completed a 5-week training protocol, which included 15 AT sessions and 12 RT sessions. A novel aspect of this protocol was that each participant had one leg performing only RT whereas the other leg performed AT+RT. Importantly, RT was conducted 6 hours after the AT session, which consisted in 40 minutes of 1-legged cycling at an intensity corresponding to 70% of the maximal power measured during an incremental test. After 40 minutes, power was increased so that participants would reach exhaustion within 1 to 5 minutes. The RT included 4 sets of 7 concentric-eccentric knee extensions on a flywheel ergometer, and participants were asked to produce a maximal effort on each repetition. During the 5 weeks of training, the power output during the RT sessions increased similarly in both legs (27 vs 28%). Partially in line with the original study conducted by Hickson³, the results also revealed that the endurance performance (time to exhaustion – TTE) was not altered by this training regimen. Actually, there was a tendency ($p=0.052$) for a time*group interaction in favour of the CT intervention (TTE improvements: +29% vs. +19%). However, this study showed that strength and power performance adaptations were similar in both legs. In addition,

Interestingly, in a meta-analysis highlighting the interference phenomenon ⁶, a theoretical model was suggested to avoid such a negative impact when athletes, because of the complex nature of their discipline, need to train for both aerobic and strength development. This model put forward the benefits of sprint training as a method to improve VO₂max ¹⁶ without compromising strength gains as RT and sprint training share similar outcomes ⁶. This hypothesis was partially verified in a study published by Cantrell and colleagues in 2014 ¹⁷. This 12-week protocol included a strength training intervention (upper and lower body, 3 sets, 4-6 repetitions, twice weekly) and a concurrent regimen, which consisted in a sprint protocol (4-6 all-out cycling efforts lasting 20 seconds each, with a 2-minute recovery period between repetitions) added to the RT intervention. Both training modes were separated by 24 hours. Whereas VO₂max was significantly improved only in the CT group, no differences were

Consequently, it appears that the interference phenomenon originally described by Hickson in 1980 ³ could be reduced or less apparent as long as training parameters are planned appropriately. Clearly, the acute organization of training sessions seems to be crucial. Indeed, training sequence, rest periods between AT and RT, and AT volume and intensity were identified as variables that could play a key role in this interference phenomenon ⁹.

However, these results were challenged by a study showing that a RT session (leg press + knee extension; 2 sets of 7 repetitions, maximal efforts on a flywheel ergometer) performed 6 hours after an AT bout (40 minutes of legged-cycling at 70% of maximal power output) did not compromise molecular responses leading to protein synthesis, whereas performances in

Taken together, the above results provide evidence that an acute bout of AT repeatedly performed with minimal rest (<15 minutes, potentially up to 24 hours) prior to a RT session could be detrimental, even in untrained participants, for optimal dynamic maximal strength development when the same muscle groups are involved in both training modes. This could be explained, at least partly, by a reduced number of repetitions performed during the RT session as a consequence of residual fatigue ¹⁹. Training status represents another potential variable that could play a role in this effect ¹¹. In addition to these observations, it seems that hypertrophy, maximal aerobic capacity and body fat percentage are not altered by the acute training sequence.

Interference between AT and RT:

- CT could be associated with compromised neuromuscular performance.
- For the moment, it is difficult to conclude on the most appropriate AT modality (running vs. cycling) as both were associated with the interference effect.

- Manipulating training variables when the main objective is related to neuromuscular performance:

- ## Side 2: The effects of concurrent training on middle- and long-distance performance

In his seminal article published in 1980 ³, Robert C. Hickson wrote: *the results of this study suggest that there is little or no benefit for endurance athletes to strength train at the same time*. While such a statement might appear quite disconnected from the knowledge acquired in the area over the last 40 years, it is important to mention that this quote mainly refers to VO₂max, which was the only aerobic performance index assessed in this research.

Since the classical work of di Prampero ²³, it is well accepted that middle- and long-distance performance is not only determined by VO₂max but also by aerobic endurance and the energy cost of locomotion (ECL). Interestingly, in a monograph publication dedicated to marathon performance, it was indicated that the latter determinant was a forgotten but crucial factor in elite running performance ²⁴. In support of this argument, it was shown that East

In 1999, an important paper was published showing that explosive strength training resulted in improvements in 5-km running performance ²⁷. During 9 weeks, a group of elite cross-country runners replaced 32% of their normal running sessions by plyometric exercises and dynamic weight training. A correlation analysis revealed that changes in ECL were associated with modifications in 5-km performance ($r = -0.54$, $p < 0.05$). Later, our research group compared plyometric exercises to dynamic weight training in a sample of recreational runners, replicating these beneficial effects on ECL with an interaction in favour of the plyometric intervention ²⁸. While these explosive strength training methods showed benefits on ECL, it is important to mention that maximal strength training could also lead to improvements in ECL ²⁹. Typically, RT interventions will result in ECL improvements ranging from 2 to 8% ³⁰.

A recent meta-analysis ³¹ suggested that a CT mesocycle was associated with improvements (net SMD = 0.52; 95%CI = 0.33 to 0.70) in middle- and long-distance performance (events > 75 seconds) in a variety of disciplines (running, cycling, cross-country skiing and swimming). Interestingly, while VO₂max and aerobic endurance (e.g. lactate or

ventilatory thresholds) were not altered by such a training intervention, it was reported that ECL was significantly enhanced (net SMD = 0.65; 95%CI = 0.32 to 0.98). In addition, training protocols with more than 24 sessions resulted in greater benefits on ECL than lower volume programs. Importantly, protocols of at least 6-8 weeks with a focus on maximal force and explosive RT were associated with greater benefits³². It was recently proposed that increased absolute strength would result in a running pattern executed at a lower (muscle contraction) relative intensity. Therefore, recruitment of higher threshold motor units would be reduced, eventually producing a more economical behaviour³³. However, the contention that reductions in ECL resulting from RT lead to improvements in middle- and long-distance performance was recently challenged³⁴. Indeed, it seems that not all studies reporting an improvement in ECL are associated with a clear performance enhancement. In addition, to our knowledge, no intervention studies allowed to establish a clear cause/effect relationship between improvements in ECL and middle- and long-distance performance. Such an observation might not be completely surprising considering that the physiological variables in the classical study by di Prampero explained approximately 72% of the performance variability in a cohort of runners from the Geneva marathon. Therefore, improvements in ECL could be unnoticed from a performance perspective considering that other factors might alter the final outcome. Nevertheless, with the available scientific literature, it appears that CT leading to a reduction in ECL might be a plausible mechanism explaining, at least partly, the observed improvements in middle- and long-distance performance. In support of this suggestion, it was shown that a reduction in ECL is accompanied by an elevation of maximal aerobic speed, even in the absence of a VO₂max enhancement²³. Therefore, such a phenomenon could help the runner achieve a faster absolute speed during a race. Besides the effects of CT on ECL, it was reported that this training modality could improve neuromuscular capacity, anaerobic metabolism and sprint performance leading to an overall improvements in middle- and long-distance

performance^{27, 35}. While it seems that less research has been conducted in that domain, one could argue that these metabolic and neuromuscular enhancements could be particularly important for specific portions of the race such as hilly sections of a course and final sprints. More precisely, it was suggested that maximal strength training is a key method to improve middle- and long-distance performance³¹. While a combination of methods (submaximal, maximal and power RT) could be associated with performance benefits, it is suggested that a long-term periodized plan should optimize training adaptations³⁶. Interestingly, as pointed out recently¹, RT methods with an emphasis on plyometric and eccentric contractions could be advantageous considering the importance of muscle fascicle length for combined sprint and endurance performance.

Manipulating training variables

While the interference phenomenon was first suggested to describe the reduction in strength gains after a CT protocol compared to a RT-only regimen^{4, 5}, it seems that CT could also lead to acute alterations on certain variables pertaining to the aerobic pathway, potentially leading to impairments in middle- and long-distance performance. Indeed, muscle damage, muscle soreness, neural fatigue and glycogen depletion induced by a RT session could acutely reduce endurance performance (time-trial, TTE and ECL). Consequently, an inappropriate training periodization repeating such negative patterns will lead to resistance training-induced sub-optimisation of endurance performance or RT-SEP³⁷. In line with this framework, a study published in 2001 revealed that an acute bout of RT was associated with significant impairments in ECL³⁸. In this study conducted with 9 distance runners (10 km time of 37:03 ± 6:09 minutes), participants had to complete 4 treadmill running sessions to assess ECL, with the first being considered as the control condition and the remaining 3 sessions implemented 1, 8 and 24 hours after a low-volume, high-intensity RT session (6 exercises, upper and lower body, 3 sets at an intensity corresponding to 8RM). When compared to the control condition,

it was shown that ECL increased by $2.6 \pm 2.3\%$ when RT was performed 1 hour prior to the treadmill running session. Importantly, ECL showed a tendency towards elevated values in comparison to the control condition up to 24 hours after the RT session. Interestingly, these impairments in ECL were accompanied by a reduction in contractile properties of the quadriceps femoris. Another study replicated a similar pattern, highlighting the acute detrimental effects of RT on ECL ³⁹. In that investigation, 14 trained runners (10 km time < 37 minutes) were randomized into 1 of the 2 following sequences: AT 6 hours after RT or the opposite. ECL and TTE tests were conducted 24 hours after the first training session (or 18 hours after the second one). Results showed that the RT-AT sequence led to impairments in ECL, which were not observed for the AT-RT sequence. Moreover, TTE performance was reduced similarly for both RT-AT and AT-RT ³⁹. Consequently, it was suggested that a recovery period of more than 24 hours should be implemented between concurrent training sessions and the following endurance performance ³⁹. Considering the above observations that aerobic and endurance performance could be acutely compromised after RT, it seems crucial to understand what could be the long-term effects of different training sequences on middle- and long-distance performance. Two recent meta-analyses concluded that the intra-session training sequence had no effect of maximal aerobic capacity (VO_{2max}) ^{21, 40}. It seems nevertheless important to highlight that VO_{2max} is definitely not the only performance factor related to middle- and long-distance events. In 2005, a study with 48 recreational athletes showed that, after a 12-week protocol with 2 sessions weekly, participants included in the AT-RT sequence improved 4 km time trial, TTE and maximal aerobic speed more than participants included in other training regimen (RT-AT sequence, AT only, RT only or a control group). While it reinforces the importance of RT as a useful stimulus for middle- and long-distance performance, it also suggests that a sequence effect could be observed, at least when considering multiple factors pertaining to endurance performance.

Training recommendations to optimize the benefits of concurrent training on middle- and long-distance performance

Middle- and long-distance athletes could benefit from CT:

- In addition to their normal training regimen, middle- and long-distance athletes (running, cycling, cross-country) could benefit from the addition of RT.
- Training should focus on the individual athlete and his/her training status.
- Maximal strength appears as a crucial training stimulus whereas plyometric/explosive training and a focus on eccentric contractions could be advantageous if implemented appropriately in the training plan.
- In a long-term periodization, it is suggested that a combination of methods (submaximal strength, maximal strength and power training) should be implemented to optimize RT adaptations.
- ECL could be improved (2-8%) whereas no significant modifications are reported for either aerobic endurance or $\text{VO}_{2\text{max}}$.
- CT interventions including more than 24 RT sessions seem particularly advantageous for the ECL.
- CT could also enhance combined performance through improvements in anaerobic metabolism and neuromuscular capacity leading to faster sprints.

Manipulating training variables when the main objective is related to middle- and long-distance performance:

- RT could acutely increase the ECL and decrease TTE for up to 24 hours and it is suggested that the accumulation of these negative effects could lead to sub-optimisation of endurance performance.
- To optimize long-term time trial performance, TTE and maximal aerobic speed, it seems that the AT-RT sequence is more beneficial than the opposite (RT-AT).

Conclusion

The combination of cardiovascular and neuromuscular solicitations within the same training program, and occasionally within the same training session, appears to be a compulsory path to achieve high performance in many sports. However, this strategy is not without risks, and may occasionally have a deleterious effect on one or more determinants of performance. The purpose of this short review was to describe the physiological adaptations induced by combined training, but also to identify the conditions that should be fulfilled to be efficient. While acute and chronic adaptations that are likely to change performance in one direction or the other are relatively well identified, it appears from this review that there is no universal rule to make sure that performance will improve. As often, success is a matter of balance. Balance in the choice of methods and their articulation within the session, the microcycle and the macrocycle, but also balance between the physiological profile of the athlete, the additional benefits that can be expected from a combined training, and the risk of injury or overreaching. Here, as for most facets of training, the key is individualization.

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